

Original article

Holocene marine intrusions in Retba and Mbawane lakes
(Senegal) evidenced by ostracod faunasMise en évidence d'intrusions marines dans les lacs Retba et Mbawane
(Sénégal) par les faunes d'ostracodes durant l'Holocène

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Abstract

The study of a sea level fluctuation based on ostracods is carried out for the first time using core sediments of Mbawane and Retba Lakes, north coast of Senegal. A total of 26 species have been identified and most of them are encountered in the Gulf of Guinea shelf and its adjacent lagoons. The Retba Lake core, which is closer to the seashore, has yielded a richer and more diversified ostracod fauna composed of four assemblages. The F1 assemblage is composed of *Cyprideis nigeriensis*, *Neomonoceratina iddoensis*, and *Aglaiochypris gambiensis*, which characterises a slightly open lagoon developing under a humid climate with its edges covered with mangrove swamps. Assemblage F2, composed mainly of *C. nigeriensis*, *N. iddoensis* and “marine species”, which is richer and more diversified, points to a marine little gulf bordered by mangrove swamps. The F3 monospecific assemblage composed of *C. nigeriensis* indicates a drying confined lagoon under an arid climate. The F4 oligospecific association at the top, composed of *C. nigeriensis* and *Pseudoconcha omatsolai*, is typical of a slightly open saline lagoon under arid climate. Core sediments from Mbawane Lake have yielded scarce and badly preserved microfaunas which are distributed into three discontinuous levels that have been correlated to those of the Retba Lake core assemblages. The F1 assemblage composed of *Leptocythere culcata* and *Xestoleberis* sp. 2 corresponds to a slightly open lagoon covered at its bottom by marine macrophytes and algae. The F2 assemblage composed of *N. iddoensis*, *Hemicytherura* sp. A, *Aglaiochypris gambiensis* and *Ruggieria tricostrata* characterises a more open lagoon bordered by mangrove swamps under a humid climate. The F3 assemblage is absent in the Mbawane Lake. The F4 monospecific association composed of *C. nigeriensis* characterises a drying lagoon under an arid climate. The overall marine influences were stronger in the Retba Lake while continental effects were dominant in Mbawane Lake during the same period. The two marine intrusions mentioned above may correspond to those of the Dakarian (3000 years B.P.) and the Saint-Louisian eras (2000–680 years B.P.) that existed during the Upper Holocene on the Senegalese coast.

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Résumé

L'étude des fluctuations du niveau marin à partir des ostracodes est menée pour la première fois le long de sondages carottés des lacs Retba et Mbawane, sur le littoral Nord du Sénégal. Vingt-six espèces ont été identifiées, dont la plupart sont connues dans la marge continentale du Golfe de Guinée et ses lagunes adjacentes. La carotte du Lac Retba, plus proche de la mer, a livré la faune d'ostracodes la plus riche et la plus diversifiée, répartie en quatre associations. L'association F1 comprend *Cyprideis nigeriensis* Omatsola, *Neomonoceratina iddoensis* et *Aglaiochypris gambiensis* qui caractérisent une lagune faiblement ouverte en climat humide et bordée de mangrove. L'association F2 dominée par *C. nigeriensis*, *N. iddoensis* et les « espèces marines » est la plus riche et la plus diversifiée et caractérise un petit golfe marin bordé de mangrove. L'association monospécifique F3 à *C. nigeriensis* caractérise une lagune confinée en voie d'assèchement en climat aride. L'association oligospécifique F4 à *C. nigeriensis* et *Pseudoconcha omatsolai* Witte au sommet de la carotte, est typique d'une lagune salée faiblement ouverte sous un climat aride. Les sédiments de la carotte du Lac Mbawane ont livré une microfaune mal conservée, répartie en trois niveaux discontinus qui ont été corrélés avec les associations du Lac Retba. L'association F1 comprend *Leptocythere culcata* et *Xestoleberis* sp. 2, espèces qui caractérisent une lagune faiblement ouverte dont le fond est recouvert d'herbes et d'algues marines. L'association F2 composée de *N. iddoensis*, *Hemicytherura* sp. A, *Aglaiochypris gambiensis* et *Ruggieria tricostrata* indique une lagune plus ouverte et bordée de mangrove sous un climat humide. L'association F3

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est absente au Lac Mbawane. L'association monospécifique F4 à *C. nigeriensis* caractérise une lagune en voie d'assèchement sous un climat aride. Dans l'ensemble, les influences marines ont été plus fortes au Lac Retba alors que les influences continentales ont prédominé au Lac Mbawane au cours de la même période. Les deux incursions marines mises en évidence pourraient correspondre à celles du Dakarien (3000 ans B.P.) et du Saint-Louisien (2000–680 ans B.P.) connues sur les côtes sénégalaises durant l'Holocène supérieur.

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Mots clés : Ostracodes ; Lagune ; Holocène ; Lac Retba ; Lac Mbawane ; Sénégal

1. Introduction

Few studies have been carried out in Senegal on ostracods from Quaternary sediments extracted from marine lagoons or estuaries. On the continental shelf, [Peypouquet \(1977\)](#), [Pinson \(1980\)](#), [Riffault \(1980\)](#), [Carbonel et al. \(1983\)](#), [Legigan et al. \(1983\)](#) have proposed a shoreline evolution model during the recent Quaternary period based on recent and fossil faunal studies. More recently, [Witte \(1993\)](#) studied shallow marine ostracods of Senegal and the Gambia. In contrast, investigations within the continent are sparse and they mainly focus on the Retba Lake and the Senegal River delta. In this latter site, the core sediments situated close to Saint-Louis bear foraminifers and ostracods which evidence a marine transgression episode around 4000 years B.P. ([Kalck, 1978](#); [Rouvillois, 1982](#)). In the Retba Lake, [Carbonel and Pinson \(1979\)](#) have reconstructed the recent closing episode of the lake from ostracod species found in core sediments (80 cm max) and proposed a post-Nouakchottian evolution model.

Recent and unpublished works on the Mbawane and Retba lakes foraminifers, ostracods and diatoms ([Sow, 2001](#); [Sarr et al., 2005](#); [Sow et al., 2006](#)) have allowed further knowledge on the climate and sea level fluctuations during the post-Nouakchottian period. The study of the shoreline evolution along the Senegalese coastline during the Quaternary has been essentially based on mollusc faunas. In the present work, the emphasis is mainly placed on the ostracod fauna for the reconstruction of the geomorphology of the Mbawane and Retba lakes, in order to highlight the role of sea level and climatic fluctuations in their post-Nouakchottian evolution.

2. Area description, methods and materials used

2.1. Geographical setting

The Retba (called also “Lac Rose”, due to its pink colour) and Mbawane lakes are located 30 and 40 km NE of Dakar respectively, along the Senegalese great coast ([Fig. 1](#)). They are separated from the sea by the “Yellow” and “White Dunes”, with an altitude between 20 and 50 m extending from Dakar to Saint-Louis. The Ogolian “Red Dunes” located eastwards are oriented NNE–SSW and border the depressions, locally called “Niayes”, where a few lakes exist. The basement of these depressions is filled with organic matter which forms the peat in the “Niayes” region. In this zone, the saline water intrusion front is maintained in equilibrium at the surface of the bottom of the “Niayes” depressions.

The Retba Lake is closer to the shoreline; a distance of 1350 m separates its feeding channel and the seashore, and the water level in the lake is 4 m below mean sea level. The surface water body is hypersaline with a salinity of about 400‰ during most of the year, except for the period after the rainy season when salinity may decrease down to 80‰ ([Sow, 2001](#)). Therefore, living faunas are almost absent in the water body, except some flagellates and halophilic bacteria which give its pink colour to the lake. Its bottom is formed by a salt crust (gypsum and halite) which covers the black coloured organic mud at the bottom.

The freshwater Mbawane Lake covers a smaller area and is located 3 km from the seashore line, where its feeding channel flows up to Kayar. Water level in the lake is approximately 1 m above mean sea level and the lake is mainly fed by rainwater and groundwater seepage. The lake is mostly dry during the year and its bottom is covered with a sandy mud.

2.2. Geological setting

The Middle Eocene basement composed of limestone, marl and clay is encountered at a depth of 35 m below the Retba Lake. It is covered by thick Quaternary sand and clay sediments. It crops out in the south of the Retba Lake and also at the east of the Mbawane Lake. The geological formations in the vicinity of these lakes are, from bottom to top, as follows ([Elouard et al., 1975, 1977](#)):

- The Ogolian (20,000–12,000 years B.P.) is represented by red dune sands which extend in the east and SE of the lakes. These dunes have been accumulated during a desert period associated with the last glacial period, when seawater level lay 120 m below the current level. From 17,000 years B.P., the sea level began to rise slowly while the climate was growing more humid.
- The Tchadian (12,000–7000 years B.P.) corresponds to a transgressive and humid episode. Sea level around 7000 years B.P. was at about the current level. The transgression movement slowed down between 7000 and 6000 years B.P., allowing the development of mangrove swamps around border lagoons. During this time, the peat accumulated in the Niayes zones while the Ogolian dunes were becoming red through the oxidation of their iron.
- The Nouakchottian (6000–5000 years B.P.) formed terraces and a talus identified at the vicinity of Retba Lake. It corresponds to the development of coquina facies combined with clayey sands deposited during a transgressive and humid

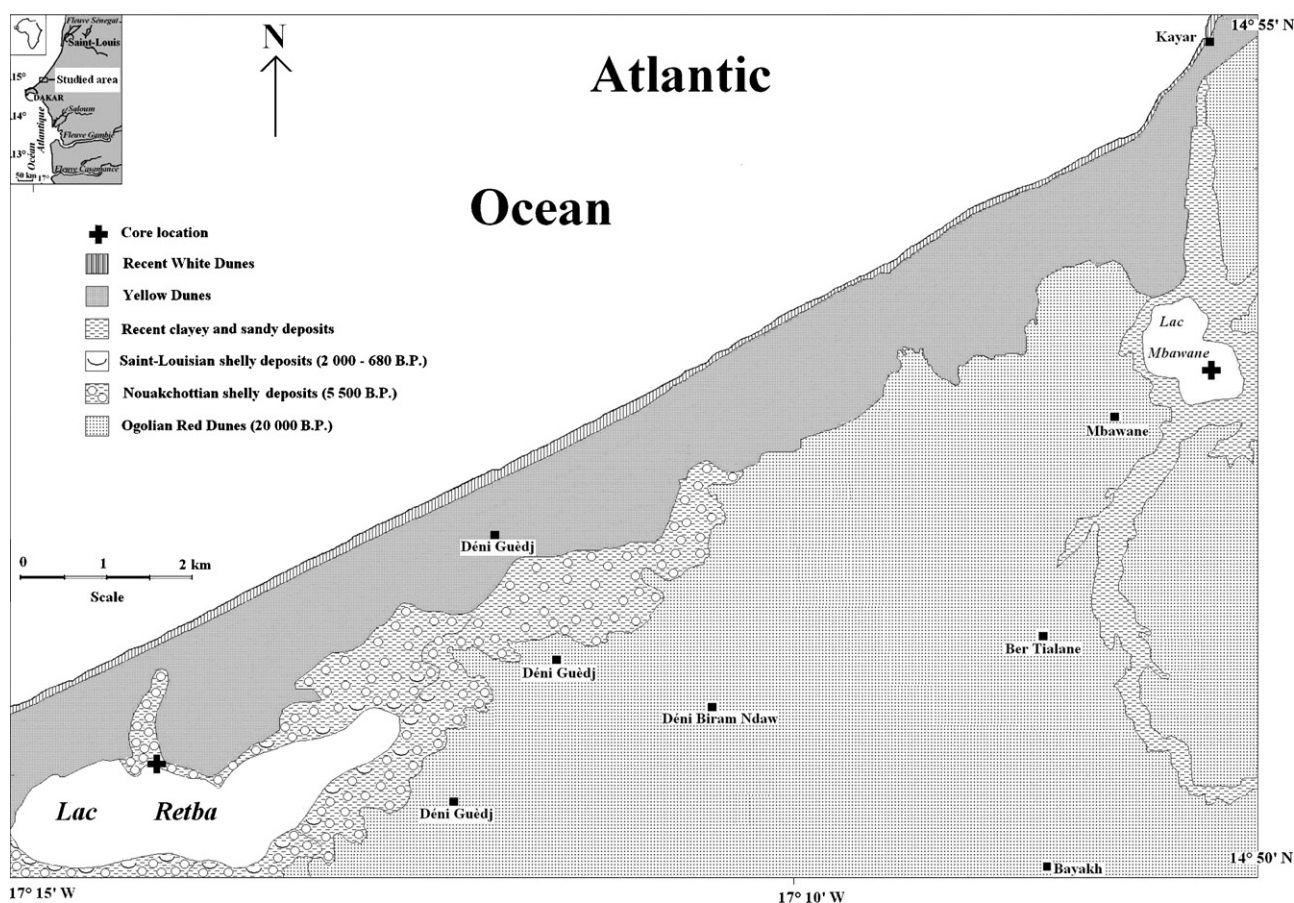


Fig. 1. Location and geological map of Retba and Mbawane lakes area.

Fig. 1. Carte de situation géographique des lacs Retba et Mbawane.

episode, when the sea had reached levels between +1 and +3 m on the Senegalo–Mauritanian littoral.

- The Tafolian (5000–2000 years B.P.) corresponds to a regressive and more arid episode, during which the littoral drift tended to regulate the coast by accumulating the sand ridges that form the “Yellow Dunes” system. This latter movement was interrupted by a marine intrusion during the Dakarian period about 3000 years B.P. and continental depressions were filled with sandy clay sediments.
- The Saint-Louisian (2000–680 years B.P.) corresponds to another marine incursion that occurred in the Retba Lake and evidenced along the Senegalese coast by the accumulation of littoral shell ridges.
- During the Recent times the littoral drift accumulated the “White Dunes” and later closed up the remaining lagoons formed during the Nouakchottian.

2.3. Material and methods

Core sediments have been obtained from two boreholes drilled in the Retba and Mbawane lakes, respectively (Fig. 1). The Retba core is 610 cm deep and was made at the sea level altitude. The Mbawane core is 600 cm deep and was made at about 1 m above sea level. Samples were collected every 10 cm,

before being dried, weighed and washed with 500 and 63 μm diameter sieves. After drying, the sediments of these sieves were then weighed. Shell, sand, silt and clay fractions have been determined. The ostracod fauna of each sample was counted, and species were identified in each sample using 10 to 40 fold stereomicroscope magnifications. We took pictures for a limited number of specimens with a Scanning Electronic Microscope at the Animal Biology Department, Faculty of Sciences and Techniques, Dakar. These data allowed us to establish the lithological sections of the cores associated with the granulometric fractions evolution in the Retba Lake, as well as the vertical distribution of the ostracods. Counting ostracods allowed to establish abundance diagrams of the main species along the cores. Ostracod samples are stored in the paleontological collection of the Department of Geology, Faculty of Sciences and Techniques, Dakar, under numbers RSO/FST/04 001 to RO/FST/04 026.

3. Results

3.1. Lithology

The core sediments are mainly clayey sands combined with a variable shell fraction. The lithological units of the cores were established from the proportions of sands, silts and clays.

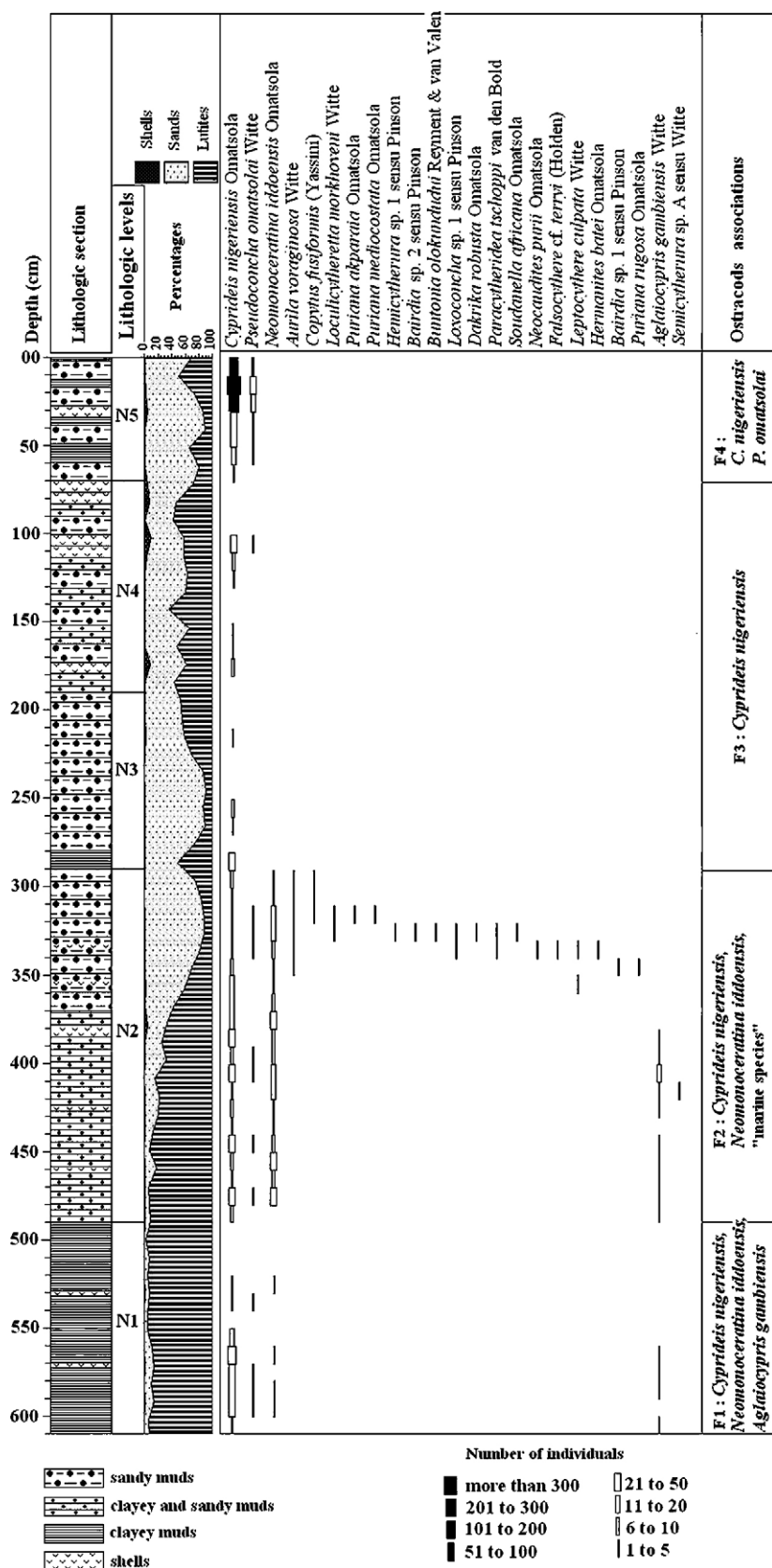


Fig. 2. Lithology and vertical distribution of ostracods in a core near the Retba Lake.

Fig. 2. Lithologie et distribution verticale des ostracodes de la carotte près du Lac Retba.

3.1.1. Retba Lake

Five lithological units identified as N1 to N5 have been identified (Fig. 2):

- N1 Unit, composed mainly of a clayey mud (610–490 cm). The proportion of sand fraction is very small and varies from 5 to 8%. The shell fraction comprises less than 1% of the sediment. It is composed of bivalves and gastropod debris, fragments of bryozoans, sea urchins, fish vertebrae, teeth, and some rare triactine sponge spicules. Organic matter and ligneous debris are abundant;
- N2 Unit, composed of sandy clay and silty mud (490–290 cm). The proportion of sand fraction increases regularly in this interval, reaching 90% at 320 cm. The shell fraction comprises less than 1% except at 370 cm where it reaches 5.7%. The faunal composition is the same as in the N1 unit and ligneous debris and organic matter are less abundant at the top of the unit. There is a minor peak of silt and clay at the top of the unit;
- N3 Unit, composed of sandy to sandy clay mud (290–190 cm). Above the peak of silt and clay, the sand fraction dominates at the bottom of this unit (290–220 cm) where it reaches 89% of the sediments. Their proportion decreases regularly to 40% at the top. In the same trend, the thin fraction increases, reaching 60% at the top (200–190 cm);
- N4 Unit, composed of sandy and clayey mud (190 to 70 cm). The sand and clay proportions change irregularly in this interval. The shell fraction is more abundant, with peaks at 180–170 cm (9.9%), 110–100 cm (10.8%) and 90–80 cm (7.9%). This latter fraction has the same faunal composition as found in the N1 and N2 units. The ligneous fragments are scarce but the sediments are richer in organic matter at some levels at the top of the unit;
- N5 Unit, composed of a sandy mud (70 to 0 cm). The proportion of sand fraction increases to 70–88% with a rich organic matter clay level between 20 and 10 cm. Shells are generally rare in this unit, but one abundance peak composed of the same fauna is noticeable at 40–30 cm (3.6%).

3.1.2. Mbawane Lake

Three units have been identified here (Fig. 3):

- N1 Unit, composed of a clayey mud (600–300 cm). The sediment is largely dominated by the clay fraction (75–90%), combined with sands (10–23%) and some sparse shell debris. The shell fraction reaches 13% at 320–310 cm interval;
- N2 Unit, composed of a sandy clay mud (300–70 cm). The sand fraction increases in the unit with richer intervals at 300–280 cm (45%) and 210–200 cm (63%). The shell fraction also increases and shows abundance peaks at 210–200 cm (25%) and 80–70 cm (50%) intervals;
- N3 Unit, mainly composed of sandy clay (70–0 cm). The proportion of sand fraction generally reaches 20% of the sediment with a maximum of 50% at the top of the core (10–0 cm). No shell fragments have been encountered in this level.

3.2. Ostracod assemblages and the living environment evolution

The vertical distribution of ostracods along the studied cores is given in Figs. 2 and 3. Figs. 4 and 5 illustrate the abundance of the main species and the faunistic density along the cores. The abundance of ostracods and their specific composition allow the distinction of four assemblages in the Retba Lake core that has a richer and more diversified fauna. This core will serve as reference to investigate the ostracod assemblages. Plate 1 illustrates the main ostracod species encountered.

3.2.1. Retba Lake

3.2.1.1. F1 assemblage (610–490 cm). It includes *Cyprideis nigeriensis*, *Neomonocerotina iddoensis*, *Aglaioocypris gambiensis* and *Pseudoconcha omatsolai*. *C. nigeriensis* dominates largely, followed by *N. iddoensis*, some *A. gambiensis* and *Pseudoconcha omatsolai* (see Appendix A). The top of the unit (520–490 cm) is ostracod free. *C. nigeriensis* mostly lives in water bodies under daily tidal influence and in confined lagoons; its predominant presence in a fauna population characterises a drying lagoon. *N. iddoensis* indicates occurrence of a desalinisation process during a humid phase and a mangrove swamp development period. Its association with the mangrove biotope has been established using the present-day fauna found in the Casamance estuary and the Senegal River delta (Rouvilleis, 1982; Debenay et al., 1990), also the Nokoué lagoon of Benin (Colleuil, 1984). It has been also established by the fossil fauna found in the continental shelf of the Cap Vert peninsula between Bargny and Mboro (Peypouquet, 1977; Pinson, 1980; Riffault, 1980). Elouard et al. (1977) and Lézine (1987) have also evidenced fossil mangrove swamps between the Cap Vert peninsula and Mauritania that had been developed in saline water bodies induced by communication between the stream fresh water and the seawater. *Aglaioocypris gambiensis* indicates shoreline proximity or an aerated sand bottom lagoon covered by algae, while *P. omatsolai* indicates salinity values close to normal marine. It is not the same bispecificism as with *Cyprideis torosa* (Jones, 1850), and *Loxoconcha elliptica* Brady, 1868, in the Mediterranean and northeast Atlantic areas. This assemblage indicates a maximum of euryhalinity or hyperhalinity (Carbonel, 1980; Callot, 1987; Fontes et al., 1985).

The F1 assemblage, therefore, characterises a slightly open lagoon bordered by mangrove swamps under tropical humid climate. The succession of rainy and dry seasons induced increasing and decreasing salt concentration in the lake.

3.2.1.2. F2 assemblage (490–290 cm). It is composed of *C. nigeriensis*, *N. iddoensis* and “marine species”. The fauna in this unit is much more diversified with 21 species identified. Between 490 and 370 cm, the F1 assemblage species are followed by *Semicytherura* sp. A. From 370 cm deep, “marine species” (20–40% of the population) identified are: *Aurila voraginosa*, *Bairdia* sp. 1, *Bairdia* sp. 2, *Buntonia olokundudui*, *Falsocythere* cf. *terryi*, *Dakrika robusta*, *Loculicytheretta morkhoveni*, *Hemicytherura* sp. 1, *Hermanites batei*, *Leptocythere culpatia*, *Loxoconcha* sp. 1, *Neocaudites purii*,

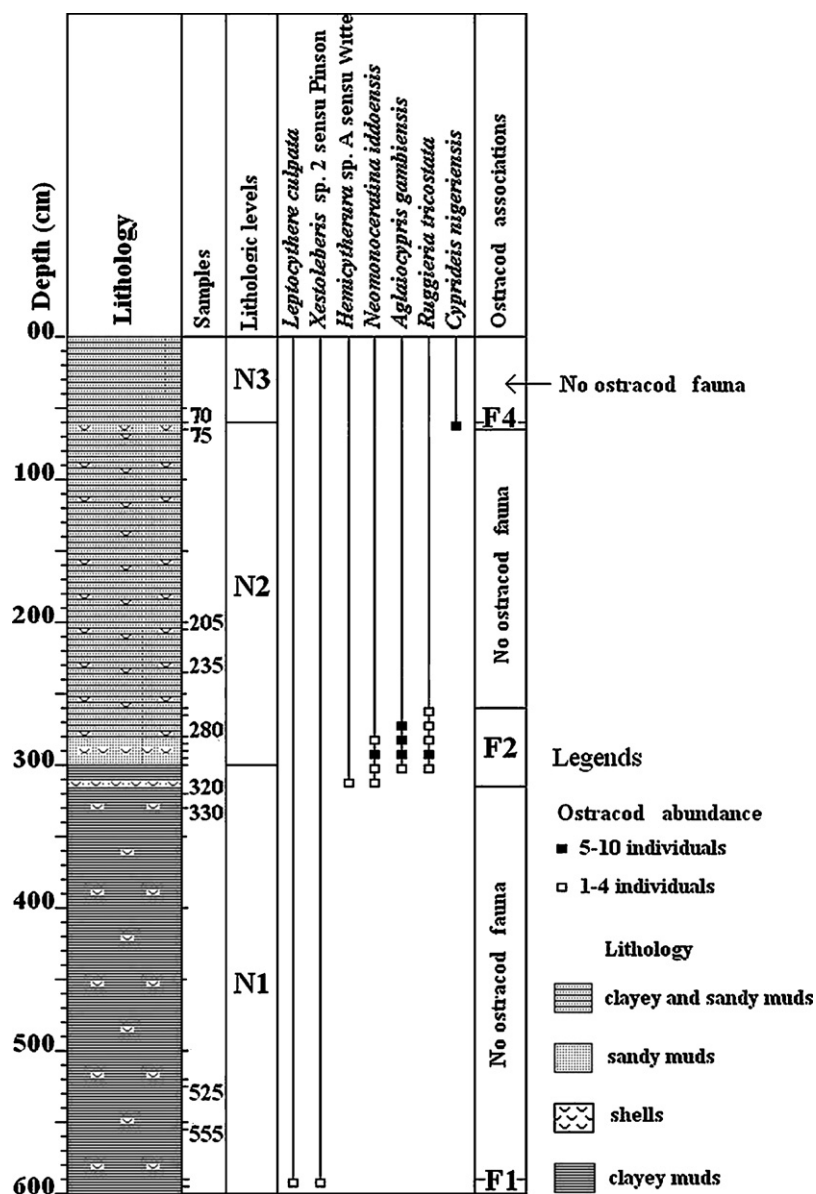


Fig. 3. Lithology and vertical distribution of ostracods in the Mbawane Lake core.

Fig. 3. Lithologie et distribution verticale des ostracodes de la carotte du Lac Mbawane.

Copypus fusiformis, *Paracytheridea tschoppi*, *Puriana akparaia*, *P. rugosa*, *P. mediocostata*, *Soudanella africana*. These species are well known and found in the Guinea Gulf (Nigeria to Ghana and Sierra Leone) and the Senegal inner shelves between 5 and 30 m below sea level (Omatsola, 1972; Pinson, 1980). The species *Bairdia* sp. 1, *Falsocythere* cf. *terryi*, *Loculicytheretta morkhoveni* and *Puriana mediocostata* belong to the shallow marine assemblage of the Senegalese continental shelf and live at water depths between 20 and 60 m (Pinson, 1980). The species *A. voraginosa*, *F. cf. terryi*, *Hemicytherura* sp. 1, *Leptocythere culpatata*, *N. purii*, *C. fusiformis*, *Paracytheridea tschoppi*, *Puriana rugosa* and *Semicytherura* sp. A, characterise the coastal phytal association living at water depths between 5 and 20 m (Pinson, 1980; Riffault, 1980).

Therefore, the occurrence of infralittoral and phytal species in this zone indicates a shallow marine environment which was

likely less than 5 m below sea level. The F2 assemblage indicates a strong marine incursion into a lagoon that had later been transformed into a small marine gulf surrounded by mangrove swamps developed under a tropical climate that was more humid than the present-day one.

3.2.1.3. F3 assemblage (290–70 cm). It is composed of scarce valves of *C. nigeriensis* often weathered and most of the interval is barren. This assemblage characterises a drying lagoon (Peypouquet, 1977; Pinson, 1980). The marine communication had been interrupted, allowing a confinement of the lagoon which became hypersaline. This saline environment favoured the development of a monospecific population of *C. nigeriensis*, which is the most resistant species both to high salinities (100–172‰) and its seasonal variations (Peypouquet, 1977; Carbonnel and Ausseil-Badié, 1985; Carbonnel, 1986; Debenay

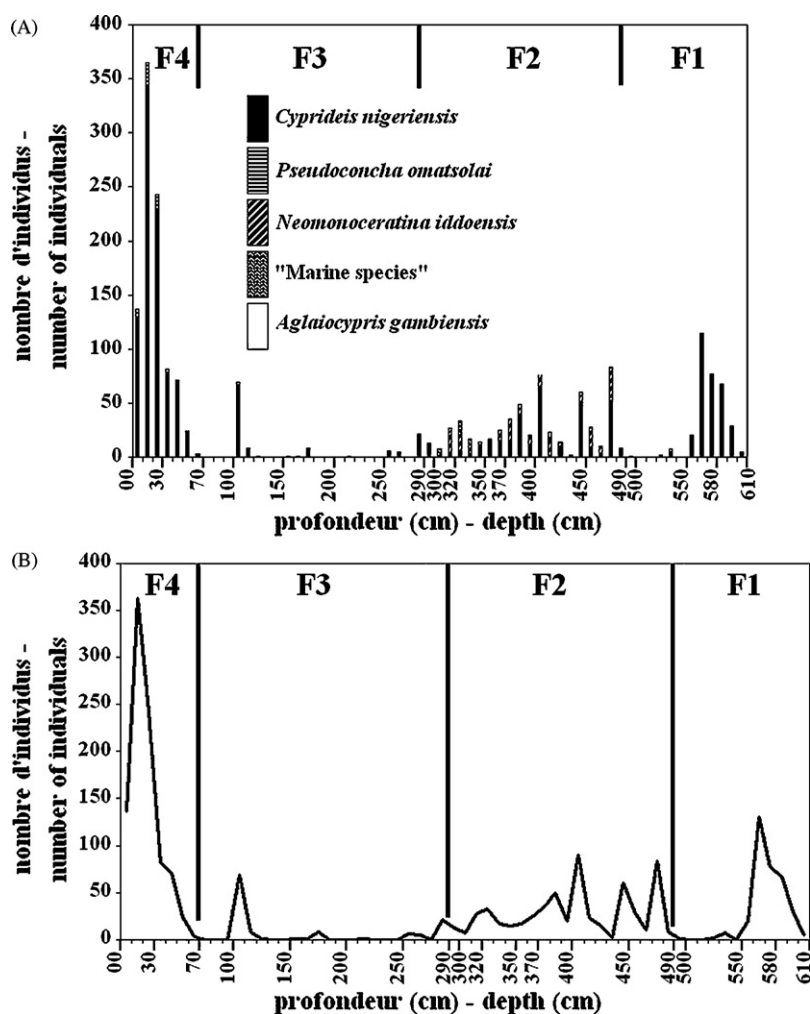


Fig. 4. Abundance of the main ostracod species (A) and faunistic density (B) in valves per 1 g sediment within the Lake Retba core.

Fig. 4. Abundance des principales espèces (A) et densité faunique (B) en valves par g de sédiment dans la carotte du Lac Retba.

et al., 1990). The absence of *N. iddoensis* may be explained by the hypersalinity of the lake induced by the arid climate, which had reduced the freshwater influx and the disappearance of the mangrove.

The F3 assemblage corresponds to the lowest seawater level recorded in the core accompanied by dune edification. The latter is the result of the littoral drift movements which have led to the lagoon confinement.

3.2.1.4. F4 assemblage (70–0 cm). It is an oligospecific assemblage composed of *C. nigeriensis* and *P. omatsolai* where *C. nigeriensis* largely dominates (>90%). The population density is particularly high at the interval between 30 and 10 cm with about 370 fossils per gram of sediment (Fig. 4B). Carbonnel (1986) revealed in his investigations in the Senegal River, the Sine-Saloum and the Casamance estuaries, that this assemblage is more developed in very saline environment.

The F4 assemblage corresponds to a less strong marine incursion under a dry climate in a slightly open lagoon. The decrease of the ostracod fauna at the top of the core (10–0 cm) probably corresponds to the latest episode of closure and confinement of the lake, which was combined with the seawater level fall.

The essential characteristics of the ostracod populations can be summarised in this description. It consists first of *C. nigeriensis* predominance in all associations. *N. iddoensis* is recorded in F1 and F2 assemblages. The "marine species" are present only at the top of F2 assemblage.

3.2.2. Mbawane Lake

Core sediments obtained from the Mbawane Lake show sparse and discontinuous ostracod faunas that correspond to F1, F2 and F4 assemblage of the Retba core (see Appendix A):

3.2.2.1. F1 assemblage (600–590 cm). It is composed of *Leptocythere culpata* and *Xestoleberis* sp. 2. These species are encountered in the coastal phytal zone and semi-confined lagoons which communicate slightly with the sea.

F1 association of the Mbawane Lake corresponds to a slightly open lagoon.

3.2.2.2. F2 assemblage (315–260 cm). It comprises four species which correspond to a mixture of lagoonal types bordered with mangrove (*N. iddoensis*) and algae covered with sandy to shelly mud marine types: *Hemicytherura* sp. A, *Aglaio-*

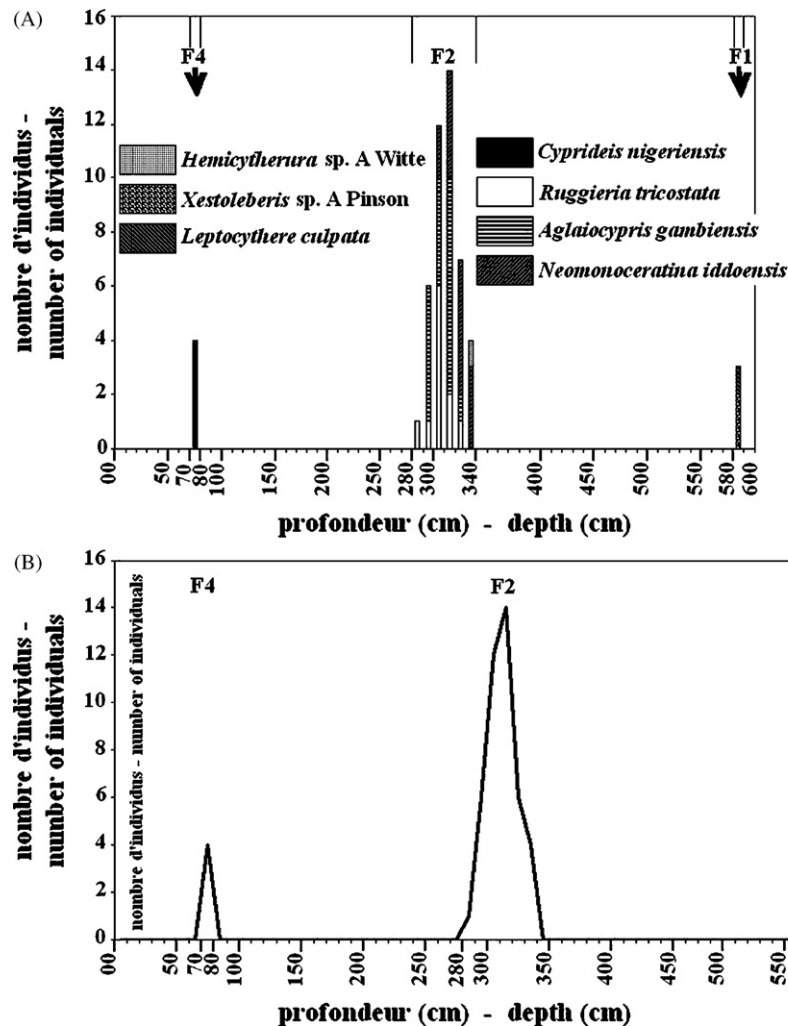
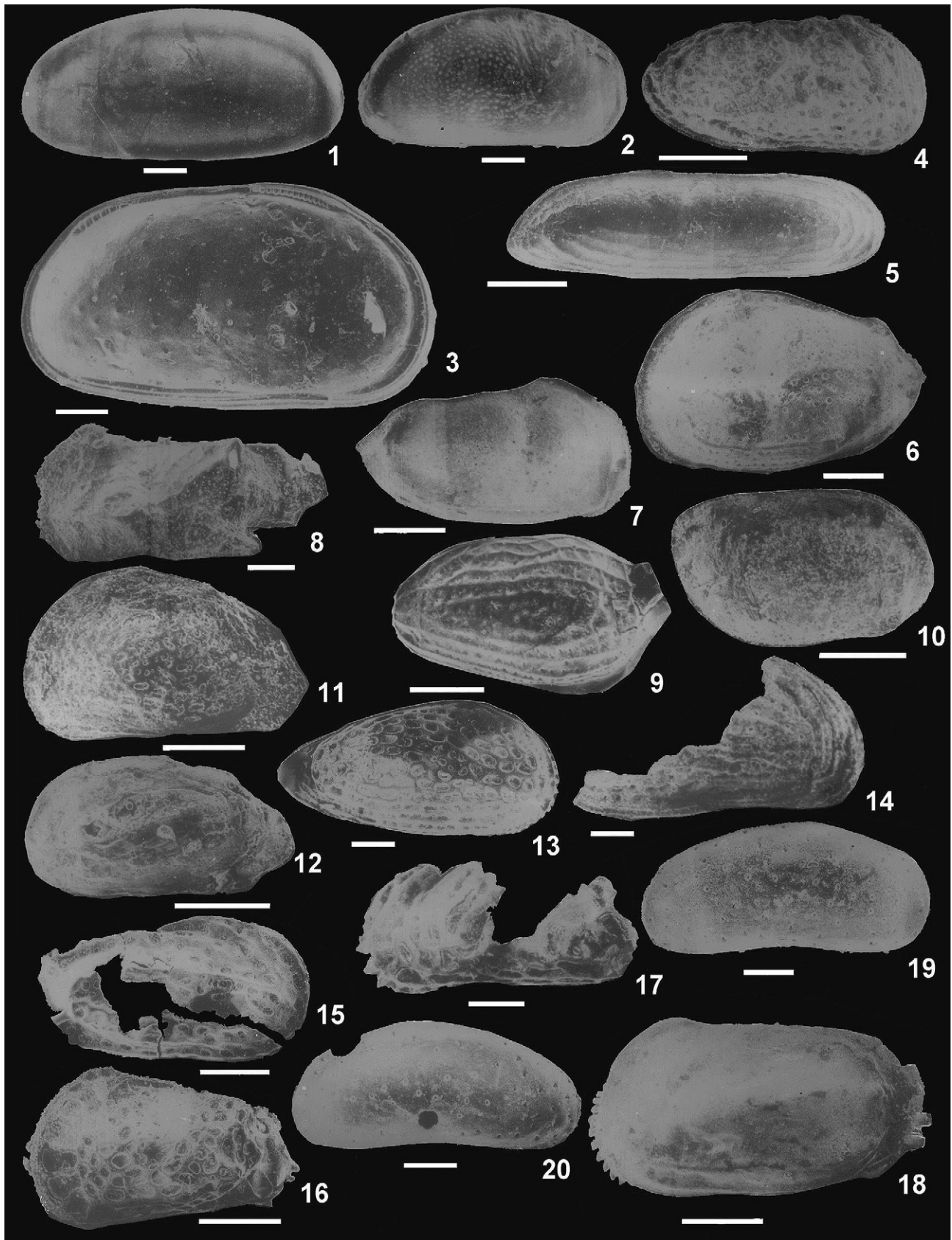


Fig. 5. Abundance of ostracod species (A) and faunistic density (B) in valves per 1 g sediment within the Mbawane Lake core.

Fig. 5. Abundance des principales espèces (A) et densité faunistique (B) en valves par g de sédiment dans la carotte du Lac Mbawane.

Plate 1. Scale bar: 100 μ m. C: carapace; RV: right valve; LV: left valve. **Fig. 1.** *Agelaiocypris gambiensis* Witte, 1993. C, left view, Retba Lake (420–410 cm), RSO/FST/04 001. **Figs. 2, 3.** *Cyprideis nigeriensis* Omatsola, 1970. 2. Punctate male RV, Retba Lake (30–40 cm). 3. Female LV, internal view, Retba Lake (50–40 cm), RSO/FST/04 002. **Fig. 4.** *Leptocythere culcata* Witte, 1993. C, right view, Retba Lake (410–400 cm), RSO/FST/04 004. **Fig. 5.** *Copypus fusiformis* (Yassini, 1979). RV, Retba Lake (320–310 cm), RSO/FST/04 003. **Figs. 6, 7.** *Neomonoceratina iddoensis* Omatsola, 1970. 6. Male C, left view, Retba Lake (410–420 cm). 7. RV, Retba Lake (380–370 cm), RSO/FST/04 005. **Fig. 8.** *Paracytheridea tschoppi* van den Bold, 1946. LV, Retba Lake (330–320 cm), RSO/FST/04 006. **Fig. 9.** *Semicytherura* sp. A Witte, 1993. C, left view, Retba Lake (420–410 cm), RSO/FST/04 007. **Fig. 10.** *Pseudoconcha omatsolai* Witte, 1993. C, right view, Retba Lake (10–0 cm), RSO/FST/04 010. **Fig. 11.** *Aurila voraginosa* Witte, 1993. C, left view (larval stage), Retba Lake (320–310 cm), RSO/FST/04 008. **Fig. 12.** *Falsocythere* cf. *terryi* (Holden, 1967). C, left view, Retba Lake (340–330 cm), RSO/FST/04 009. **Fig. 13.** *Buntonia olokundui* Reymont and van Valen, 1969. C, right view, Retba Lake (320–310 cm), RSO/FST/04 011. **Fig. 14.** *Soudanella africana* Omatsola, 1972. Partially dissolved fragment, RV, Retba Lake (330–320 cm), RSO/FST/04 012. **Fig. 15.** *Loculicytheretta morkhoveni* Witte, 1986. Partially dissolved C, right view, Retba Lake (330–320 cm), RSO/FST/04 013. **Fig. 16.** *Hermanites batei* Omatsola, 1972. C, left view, Retba Lake (340–330 cm), RSO/FST/04 014. **Fig. 17.** *Puriana rugosa* Omatsola, 1972. Partially dissolved LV, Retba Lake (350–340 cm), RSO/FST/04 015. **Fig. 18.** *Ruggieria tricostata* Omatsola, 1972. LV, Mbawane Lake (265–260 cm), RSO/FST/04 016. **Figs. 19, 20.** *Xestoleberis* sp. 2 Pinson, 1980. 19. RV. 20. Partially dissolved LV, Mbawane Lake (595–590 cm), RSO/FST/04 017.

Planche 1. Barre d'échelle: 100 μ m. C: carapace; VD: valve droite; VG: valve gauche. **Fig. 1.** *Agelaiocypris gambiensis* Witte, 1993. C, vue droite, Lac Retba (420–410 cm), RSO/FST/04 001. **Figs. 2, 3.** *Cyprideis nigeriensis* Omatsola, 1970. 2. Mâle ponctué, VD, Lac Retba (30–40 cm). 3. Femelle, VG, vue interne, Lac Retba (50–40 cm), RSO/FST/04 002. **Fig. 4.** *Leptocythere culcata* Witte, 1993. C, vue droite, Lac Retba (410–400 cm), RSO/FST/04 004. **Fig. 5.** *Copypus fusiformis* (Yassini, 1979). VD, Lac Retba (320–310 cm), RSO/FST/04 003. **Figs. 6, 7.** *Neomonoceratina iddoensis* Omatsola, 1970. 6. C, mâle, vue gauche, Lac Retba (410–420 cm). 7. VD, Lac Retba (380–370 cm), RSO/FST/04 005. **Fig. 8.** *Paracytheridea tschoppi* van den Bold, 1946. VG, Lac Retba (330–320 cm), RSO/FST/04 006. **Fig. 9.** *Semicytherura* sp. A Witte, 1993. C, vue gauche, Lac Retba (420–410 cm), RSO/FST/04 007. **Fig. 10.** *Pseudoconcha omatsolai* Witte, 1993. C, vue droite, Lac Retba (10–0 cm), RSO/FST/04 010. **Fig. 11.** *Aurila voraginosa* Witte, 1993. C, vue gauche (stade larvaire), Lac Retba (320–310 cm), RSO/FST/04 008. **Fig. 12.** *Falsocythere* cf. *terryi* (Holden, 1967). C, vue gauche, Lac Retba (340–330 cm), RSO/FST/04 009. **Fig. 13.** *Buntonia olokundui* Reymont et van Valen, 1969. C, vue droite, Lac Retba (320–310 cm), RSO/FST/04 011. **Fig. 14.** *Soudanella africana* Omatsola, 1972. Fragment partiellement dissout, VD, Lac Retba (330–320 cm), RSO/FST/04 012. **Fig. 15.** *Loculicytheretta morkhoveni* Witte, 1986. C, partiellement dissoute, vue droite, Lac Retba (330–320 cm), RSO/FST/04 013. **Fig. 16.** *Hermanites batei* Omatsola, 1972. C, vue gauche, Lac Retba (340–330 cm), RSO/FST/04 014. **Fig. 17.** *Puriana rugosa* Omatsola, 1972. VG, partiellement dissoute, Lac Retba (350–340 cm), RSO/FST/04 015. **Fig. 18.** *Ruggieria tricostata* Omatsola, 1972. VG, Lac Mbawane (265–260 cm), RSO/FST/04 016. **Figs. 19, 20.** *Xestoleberis* sp. 2 Pinson, 1980. 19. VD. 20. VG, partiellement dissoute, Lac Mbawane (595–590 cm), RSO/FST/04 017.



cypris gambiensis and *Ruggieria tricostata* (Omatsola, 1972; Pinson, 1980; Riffault, 1980).

The F2 assemblage reflects here a more open lagoon surrounded by mangrove swamps under a tropical humid climate.

3.2.2.3. F4 assemblage (80–70 cm). This monospecific *C. nigeriensis* assemblage is featured by badly preserved valves due to dissolution processes, and characterises a drying confined lagoon (Peypouquet, 1977; Rouvillois, 1982).

The F4 assemblage corresponds to a low sea level, confined lagoon and dry climate.

3.2.2.4. F3 assemblage. The F3 assemblage of the Retba core is absent in the Mbawane core. Ostracod populations are also discontinuous with barren levels (intervals 590–320 cm, 260–80 cm and 70–0 cm) in the core. In intervals 590–320 cm and 260–80 cm, the absence of ostracods is probably due to the unfavourable preservation conditions which were characterized by organic matter, rich sediments and an acidic environment. The occurrence of partially dissolved foraminiferal tests evidences it. At the top core (70–0 cm), the absence of microfauna is mainly due to the lagoon confinement conditions which had evolved into a continental environment characterised by a predominantly seasonal freshwater influx. The presence of freshwater diatom species confirms the latter.

4. Evolution of the Retba and Mbawane lakes

The ostracod populations, particularly the one encountered in the Retba Lake, allows the sketching of an evolution model of these lakes during Upper Holocene times (Fig. 6). Previous investigations reveal that the studied sediments from this lake lie on the Nouakchottian (5500 years B.P.) beach shell deposits. In this study, despite the lack of age determination of the core sediments, we will attempt to reconstruct the post-Nouakchottian marine incursions that occurred in the Retba Lake and the Senegalo–Mauritanian littoral (Elouard et al., 1975, 1977).

4.1. Slightly open lagoons (F1 assemblage)

During the first stage evolution found in the bottom core sediments, the lagoons were slightly open to the sea and were receiving a significant freshwater influx derived from local precipitations and groundwater seepage under a more humid climate. The latter had favoured a mangrove swamp development. The microfauna disappeared in the upper part of the assemblage—a feature which occurred earlier in the Mbawane Lake than in the Retba one—which may have many causes:

- a closure of the marine communication through a littoral band which had led to a decreased water level;
- the development of a vegetation and hence organic matter production which tended to weather the tests and shells.

The dissolved tests of foraminifers observed in the level combined with thin sediments favour the filling up of the lagoons by sediments and vegetation development. Progressive filling up of

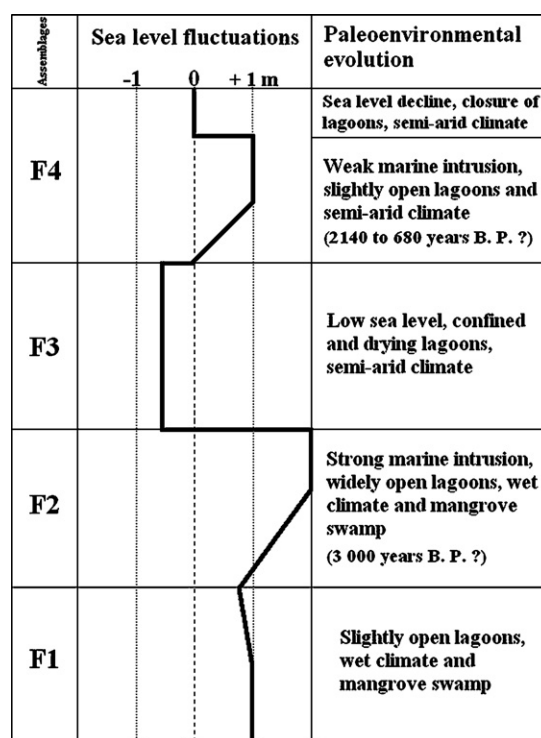


Fig. 6. Sea level and paleoenvironmental fluctuations of the Retba and Mbawane lakes during the Upper Holocene.

Fig. 6. Fluctuations du niveau de la mer et du paléoenvironnement des lacs Retba et Mbawane pendant l'Holocène supérieur.

lagoons during the Recent Quaternary has been also reported by Debenay and Bellion (1983) in the south of Dakar and may have occurred at the end of the Tafolian regression around 4000 years B.P.

4.2. Strong marine intrusion, widely open lagoons and humid climate (F2 assemblage)

The second stage (F2) corresponds to lagoons which have been more intruded by seawater. This phenomenon appeared earlier and was stronger in the Retba Lake which had been transformed into a small marine gulf. In contrast, the Mbawane Lake grew into an open littoral lagoon evidenced by the high sand and mollusc shell proportion. This marine incursion combined with a humid climate could be associated with that of the Dakarian (3000 years B.P.) when sea level rose to 2 m above the mean sea level (Elouard et al., 1977). This is in accordance with the higher sea level recorded in F2 assemblage compared to F4 assemblage. This hypothesis gives a more convincing explanation for the aridity of the climate noticed since the F3 assemblage. In fact, the aridity trend of the Sahel climate, according to estimates, dates back after 3000 years B.P. (Elouard et al., 1975; Carbonel and Pinson, 1979; Lézine, 1987).

4.3. Low sea level, confined and drying lagoons (F3 assemblage)

The third stage (F3) corresponds to the closure of the lagoons feeding channels induced by the decrease in seawater level and

the arid climate. This has led to the development of a monospecific population of *C. nigeriensis* in the Retba Lake. In contrast, at Mbawane Lake, the absence of this assemblage may be due to the longer distance from the sea and the higher altitude of the lake which had led to its confinement and drying-up. This level corresponds to the lowest seawater level revealed by the core sediments and may be lower than the present-day one. The littoral drift had allowed the formation of dunes, which had narrowed the feeding channels, while confining the lagoons. Therefore, the reduced freshwater influx tended to increase the aridity of the local climate, the evaporation and increasing salt concentration in the lakes. As a consequence, the lake bordering mangroves disappeared.

4.4. Weak marine intrusion, slightly open lagoons, semi-arid climate (F4 assemblage)

The fourth stage evolution (F4) corresponds to a weak sea level rise. During this period, the Retba Lake became a slightly open saline lagoon where the oligospecific population of *C. nigeriensis* and *P. omatsolai* developed, while the Mbawane Lake formed a drying lake under semi-arid climate with development of monospecific population of scarce *C. nigeriensis* (Peypouquet, 1977). This sea level rise may correspond to the more recent marine incursion in the Retba Lake that has been dated by Elouard et al. (1975) between 2140 and 680 years B.P. This transgression corresponds to that of the Saint-Louisian defined in the Senegalo–Mauritanian basin regional stratigraphic scale during which the sea level rose +1 m above the present-day sea level.

4.5. Sea level fall, closure of the lagoons and semi-arid climate (top of F4 assemblage)

The last stage is not well recorded by the ostracod fauna and extends into the upper part of the cores. It reveals a definite confinement of the lakes which was the consequence of the sea level fall and semi-arid climate. This is evidenced in the Retba Lake feeding channel by the poor ostracod population, the maintained oligospecific assemblage, while salt crust accumulated at the bottom of the lake (Carbonel and Pinson, 1979). In contrast, the Mbawane Lake functioned like a collecting basin which was receiving rain waters and groundwater seepage influx.

From these results, it is clear that marine influences were stronger in the Retba Lake than in the Mbawane which was predominantly subjected to continental influences. In the latter environment, the dominantly freshwater influx had negative effects on both the development and preservation of the ostracods.

Yet, these results and their interpretation need confirmation through ^{14}C datings.

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Appendix A

List of Upper Holocene ostracod species of the Mbawane and Retba lakes

Agelaiocypris gambiensis Witte, 1993
Aurila voraginosa Witte, 1993
Bairdia sp. 1 Pinson, 1980
Bairdia sp. 2 Pinson, 1980
Buntonia olokundudui Reymont and van Valen, 1969
Copypus fusiformis (Yassini, 1979)
Cyprideis nigeriensis Omatsola, 1970
Dakrika robusta Omatsola, 1972
Falsocythere cf. *terryi* (Holden, 1967)
Hemicytherura sp. 1 Pinson, 1980
Hemicytherura sp. A Witte, 1993
Hermanites batei Omatsola, 1972
Leptocythere culpata Witte, 1993
Loculicytheretta morkhoveni Witte, 1986
Loxoconcha sp. 1 Pinson, 1980
Neocaudites purii Omatsola, 1972
Neomonoceratina iddoensis Omatsola, 1970
Paracytheridea tschoppi van den Bold, 1946
Puriana akparaia Omatsola, 1972
Puriana mediocostata Omatsola, 1972
Puriana rugosa Omatsola, 1972
Pseudoconcha omatsolai Witte, 1993
Ruggieria tricostata Omatsola, 1972
Semicytherura sp. A Witte, 1993
Soudanella africana Omatsola, 1972
Xestoleberis sp. 2 Pinson, 1980

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